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Appendix

FORM 1

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
Diploma in Software Engineering

Programme: DSFS1Y1 (Group: 2)

Assignment

AMSE1003 SOFTWARE ENGINEERING

Name (Block Letters)	Registration No.	Signature	Marks
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Date of Submission:

FORM 2

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

Plagiarism Statement and Guideline for Late Submission of Coursework

Read, complete, and sign this statement to be submitted with the written report.

We confirm that the submitted works are all our own work and are in our own words.

Name (Block Letters)	Registration No.	Signature	Date
1.			
2.			
3.			
4.			
5.			

Main report

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Location: TAR UMT

Tunku Abdul Rahman University of Management and Technology(TAR UMT) is a leading prominent private institution of higher education in Malaysia.Its main campus is located at Jalan Genting Kelang, Setapak, Kuala Lumpur, where it serves a large community of students and staff.As a comprehensive university, TAR UMT provides a wide range of academic programmes spanning from pre-university to postgraduate levels.

To support daily campus operations, the university's Students Affairs Division manages various administrative services, including the lost and found facility. Currently, these operations are handled through manual, paper-based logging and informal communication channels like Whatsapp.Because of the high numbers of students on campus, this manual process has become highly inefficient, creating administrative challenges such as disorganization, lack of security, and data loss.This highlights the urgent need to transition to a centralized, digital **Lost and Found Item Management System** designed to systematically **record, search, and manage lost and found items within university premises.**

Part 1

Part 1.1 Problems of existing systems

1. **Unorganized Systems**
   Lost items in the system are unsorted and disoriented so users will have difficulty finding a certain object or item inside the unorganised system.

2. **Security risk**
   Lost items do not guarantee accurate owners and it can be claimed by anyone on the third floor on block A , because the cabinet is not an enclosed rack. Furthermore, there is no admin to manage those lost items. Admin shall have a dedicated key to ensure the cabinet will not be opened by anyone to claim items without any permission.

3. **No Platform for the person to look for their lost items**
   Students and staff must visit the lost and found office in person to identify whether a found item belongs to them.

4. **Data Loss Risk**
   Staff members responsible for handling lost and found items rely on manual, paper-based logs and disconnected spreadsheets to record item details, dates, and claimant information. This ad hoc system was likely adequate when item volume was low, but it creates several compounding issues as operations grow.
   Physical paper logs can be misplaced, damaged (spills, tears, fading ink), or discarded accidentally. Spreadsheets stored locally on individual computers or USB drives can be lost due to hardware failure, accidental deletion, or lack of backups.

5. **Lack of centralization for lost and found information**
   - Information that is spread across notice boards, security desks, and front desks will use paper logs and be scattered in different departments. This will prevent real-time tracking, several delays for matching the lost and found items, and significantly lower the chances of the owner to recover their belongings.

Part 1.2 Software quality attributes of the project

1. **Efficiency**

- Efficiency refers to the ability of the system to perform tasks quickly while minimizing the system resources.
  -The system should have real time notification mechanisms, thus when someone finds an item and logs it in, the system rings automatic updates about the lost claims.

2. **Acceptability**

- Acceptability refers to the system being user-friendly and it is easy to understand,as well as suitable to use by intended users.
  -The system should be easy to navigate so users can quickly report on lost items or search for the owner even with minimal technical skill.

3. **Dependability & Security**

- Dependability & Security refers to the system operating reliably and consistently while protecting data from unauthorized access.
  -The system should ensure that only authorized admin can modify item records,while the users should only be able to search,report and claim items through the system.

4. **Maintainability**

- Maintainability refers to the ability of the system to be modified,updated or improved when there is change in the requirement.
  -The system should ensure that staff members can easily record and organize lost items so that it doesn't go missing.

5. **Reliability and availability**

- Reliability and availability refers to the ability of the system to operate consistently and remain accessible whenever users need it.

  -The platform needs high uptime around 99.9% so users can search or report items 24/7, and with data backed up to prevent loss in case of server failure.

Part 1.3 Software Process Model

The software process model chosen is Agile: Scrum.

Scrum is an Agile software development methodology based on iterative and incremental development,where a project is completed through a series of short development cycles called “sprint”,the duration for these cycles usually around 2 - 4 weeks.It focuses on collaboration among team members which includes junior developer and experienced experts,continuous feedback,and ability to adapt to changing requirements.Therefore,functional features can be delivered quickly and improved throughout the project.

This software process model is suitable for this project,due to the fact that the agile scrum system offers a process to break it into small pieces and incremental cycles called sprints that is typically 1 to 2 weeks long. Each sprint results in a working, testable software.It is assumed that the project team needs to hire experts with sufficient knowledge and experience in Agile Scrum to manage sprint planning,development,and testing efficiently.

- Sprint 1 (Reporting Module)

Focuses on the basics, the base to log a lost item and to log a found item,establishing the initial database entries.

- Sprint 2 (Search Module)

The introduction to filtering categories, the availability to search items and the basic user authentication.

- Sprint 3 (Lost Item Management)

Focuses on physical inventory tracking.allowing admins to distribute items,allowing admins to map found items to specific physical storage locations like shelves and bins.

- Sprint 4 (Claim verification and Updates)

A function that provides admins with access control to secure user-submitted items,enabling them to verify proof of ownership such as photos and update claim statuses.

- Sprint 5 (User Management and Role-Based Security)

Focuses on finalizing secure registration, login, email OTP verification, and locking down the system with strict role-based access control for standard users and administrators.

Justification for selecting Scrum as the software process system:

1. **Fast delivery of a minimum product**
   - Sprint cycles allows short development of system features such as user login and items searching to be developed and delivered early,allowing the system to be used sooner.

2. **Feedback usable for users**
   - At the end of each sprint,students and staff can test the current version of the system to provide feedback for improvement.Their suggestions can be used to improve existing features and to add new features.

3. **High adaptability to secure and privacy standards**
   - Scrum makes it easy to improve security features throughout the project.For instance,user login and admin authorization can be updated in different sprints to ensure only authorized users can modify records.

4. **Mitigates risks for complicated features**
   - Development of the system in small stages can reduce the risk of errors.Complex features such as automatic notification can be developed,tested and fixed before moving onto the next features.

Part 2

Part 2.1 Project Plan and Schedule

	Duration (Week)											
Activities	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Product Backlog												
Planning												
Develop Reporting Module												
Sprint Backlog												
Sprint Planning												
Sprint review 1												
Develop Search Module												
Sprint Backlog												
Sprint Planning												
Sprint review 2												
Develop Lost Item Manageme nt												
Sprint Backlog												
Sprint Planning												

Sprint review 3												
Develop Claim verification and Updates Module												
Sprint backlog												
Sprint Planning												
Sprint review 4												
Develop User management and authentication Module												
Sprint Backlog												
Sprint Planning												
Sprint review 5												

Part 2.2 Software Requirements Specification

System Modules:

- user management and authentication: registration and login, role based control (standard user, staff, finder, admin). (Randolf)
- Reporting module: lost item reports, found item logs. (Elisha)
- Search system: filtering categories and searches.(Ivan)
- Claim verification and updates: evidence of ownership, real time notifications(Elias)
- Lost Item Management: organizing physical item locations and status(kevin)

1.User Management and Authentication(Randy)

General description: Authentication handles identity verification, while Role-Based Access Control (RBAC) dictates what actions a verified user can perform within the system.

Registration: New users sign up using an email address, phone number, or third-party Accounts. During sign-up, users provide basic profile information like name, contact information and accept privacy terms. Email or SMS verification (OTP) is required to prevent fake accounts.

Login & Session Management: Registered users log in with their credentials.

Account Recovery & Security: Includes self-service password reset via verification links, options for Multi-Factor Authentication (MFA), and automated account lockout after multiple failed login attempts.

Functional requirement

1.1.The system shall be able to authenticate users
1.2.The system shall be able to register with emails and accounts to access the system
1.3The system shall be able to edit profiles, contact information, usernames inside the system.
1.4The system shall be able to send an email OTP for security.

Non-functional requirement

1.1The system shall be able to send the OTP within 3-10 seconds.

2.Reporting Module(Eli)

General description: The Reporting Module allows students and staff to manage,record and track lost or found items systematically.Students and staff can write descriptions of the items that they lost as well as include photos of the item to make it easier to find their lost item. It also allows users to edit submitted lost items reports when there is new information available to be added or changed.This module is able to detect any duplicate reports entries to prevent overlapping reports and ensure accurate tracking of all lost and found items.

Functional requirement
2.1The system shall allow users to submit lost item reports.
2.2The system shall allow users to edit submitted reports.
2.3The system shall allow users to upload photos with reports.
2.4The system shall be able to detect duplicate reports.

Non-functional requirement
2.1The system shall process lost item reports within 1 minute.

3.Search system(Ivan)
General description: The Search System module allows students, staff, and lecturers to locate lost or found items that have been logged into the system without needing to visit the admin or staff office in person. Users can search by keywords such as item name, brand, colour and narrow the results using filters such as category, date reported, and location found. The module retrieves matching records from the Reporting Module's database and displays them in an organized, easy to read list so users can quickly identify if their item has been found.

Functional requirement
3.1.The system shall allow users to search for items using keywords such as item name, brand, or colour.
3.2.The system shall allow users to filter search results by category such as electronics, clothing, documents, accessories.
3.3.The system shall allow users to filter search results by date range , date lost/found and location such as which block or floor.
3.4.The system shall display search results as a list showing item description, category, date, and status.
Non-functional requirement
3.1.The system shall return search results within 3 seconds under normal load, to align with the Efficiency quality attribute.

4.Claim verification and updates(Elias)
General description:
The claim verification system shall allow admin to check the system's item claim history, and do updates about the lost item if the lost item had been claimed by someone. The system also shall allow admin to reject the application by user if the proof given is not accurate. Plus, the system shall allow the user to upload a photo as a proof that the lost item belongs to him or her. Lastly, the system shall have encrypted user data as a protection to prevent cyber criminal activities.

Functional requirement
4.1.The system shall allow the user to provide and upload photos to prove that the item belongs to him.For example receipt and the unique feature of the item.
4.2.The system shall allow admin to do updates about the status of claim , give notification automatically to claimants like ‘accepted ‘ , ‘ rejected’ , and ‘ claimed’.
4.3.The system shall allow admin to verify the details of lost item for each proof photo that was provided by claimants.
4.4.The system shall have a claim history of all claims that have been made , so that a complete audit trail is available if disputes arise or if multiple people claim the same item.

Non-functional requirement
4.1.The system shall encrypt all personal data and photo proof that are submitted , and only authorized people are allowed to check the data .

5.Lost Item Management (Kevin)
General description: Admin are in charge of keeping track of the status of the lost item whether it has been claimed or still in the lost and found storage. The number of lost items found are then divided into different categories and are also divided into different alphabet letters based on the first letter of the item's names.

Functional requirement
5.1.The system allows admins to enter item details like category, description, and date found.
5.2.The system allows Admins to assign a specific shelf, bin, or room number to each item based on the category.
5.3.The system allows admins to change an item's status whether it has been claimed or not.
5.4.The system allows admins to look up items using keywords or ID numbers.
Non-functional requirement
5.1.The system shall only allow authorized admin accounts to view, edit, or delete inventory records.

Part 2.3 An architectural design

| [register / login]
| [main menu / exit]
| [report lost / found item]
| [data base/ file storage]

| [main menu / exit]
\/

1. Register / Login Layer — Ivan

Role: Authenticates users and determines their access role (user, staff, or admin) before granting entry to the system.

- Allows new users to register an account
- Allows returning users to log back in
- Verifies role/permissions so later modules know what each user can see and do
- Presents a clean, polished interface as the system's entry point

2. Main Menu — Randy

Role: allows users, staff members and admins to choose between reporting a lost or found item, or exit the system from the optional-displaying main menu screen and to return back to the main menu after exiting a module.

- Displays available options based on the logged-in user's role
- Routes users to the Report module or the Database module
- Receives control back from either module once a task is finished
- Handles Exit — the only point where the program actually terminates

3. Report Lost or Found Item — Elias

Role: allow users to do reports about lost or found items, to take photos and upload to be as a proof. Allow staff members and admin to classify each lost and found item following each item's categories.

- Users: submit a lost or found item report, attach a photo as proof
- Staff/Admin: review submitted reports and classify each item by category
- Passes classified reports on to the Database module for storage

4. Database / File Storage — Kevin

Role: Allow staff to add, update and delete lost items reported with description and photo as evidence. Staff can organize the lost items based on their description, so staff can find the item easily to return back to the owner. Students can check any lost item that belongs to them and then staff can see if it is in the storage.

- Staff/Admin: add, update, and delete stored item records (description + photo evidence)
- Staff/Admin: organize items by category/description for fast retrieval
- Users: search/check whether an item matching their description has been found
- Staff: cross-reference a match against user reports to confirm return of the item

Part 3

Part 3.1 Test cases

User management and authentication Module

No	Objective	Test Data	Expected Result	Actual Result	Remarks/ Comments
1.	To login using a valid account (student's email)	User's name "User's name" User's account "User's email account" User's password "User's email account password"	Display successful authentication		
2.	To login using an invalid account (student's email)	User's name "User's name" (wrong input) User's account "User's email account" (wrong input) User's password "User's email account password" (wrong input)	Display unsuccessful authentication due to (name/account/password)		
3.	To edit profile picture and information using appropriate words and images	To change profile photo "add photo" To add/change information "add bio"	Display "saved successful changes"		
4.	To edit profile picture and	To change	Display "saved		

	information using inappropriate words and images	profile photo "add photo" (invalid input) To add/change information "add bio" (invalid input)	unsuccessful changes please change"		
5.	To send automatic OTP pins to register an email account (student emails)	The system shall send a randomly regenerated OTP pin to the user's account input	Display a randomly generated OTP pin number sended to the user's email address		

Reporting Module

No	Objective	Test Data	Expected Result	Actual Result	Remarks/ Comments
1.	To submit lost item reports using valid data	Description "A black wallet" Category "Personal belonging" Reporter name "Ali"	The lost item reports recorded successfully.		
2.	To submit lost item reports using invalid data	Reporter name "NULL"	The lost item reports recorded unsuccessfully.		
3.	To upload a valid photo format with a report	File:lost_purse.jpg Format:jpg	The photo uploads successfully and attaches to reports.		
4.	To upload an invalid photo format with a report	File:lost_purse.pdf Format:pdf	The photo uploads unsuccessfully and does not attach to reports.		

5.	To detect duplicate lost item reports using valid data	Report 1 Reporter name “Ali” Item name “iPhone 13” Category “Electronics” Report 2 Reporter name “Ali” Item name “iPhone 13” Category “Electronics”	The second report is blocked and the system will alert the user with the message “Duplicate report detected!”		

Search Module

No	Objective	Test Data	Expected Result	Actual Result	Remarks/ Comments
1.	Search using valid keyword data	Keyword: "black wallet”	System displays matching items with description, category, date,and status		
2.	Search using invalid/non-existent data	Keyword: “gun”	System displays "No items found" message		
3.	Filter search results by category using valid data	Category: "Clothings”	Only items tagged “Clothings”are displayed		
4.	Filter search results by date range using valid data	Date range: 01/08/2026 – 20/08/2026	Only items reported within this range are shown		
5.	Filter by location (block/floor) using valid data	Location: "Block A, Level 2”	Only items matching that location are displayed		

Claim verification and updates Module

No	Objective	Test Data	Expected Result	Actual Result	Remarks/ Comments
1.	To claim a lost item by using valid data	Item name : “Black wallet” Lost date : 12/12/2026 Location lost: canteen	Claims successfully, and claim status is updated.		
2.	To claim a lost using invalid data	Item name : “White wallet” Lost date : 13/12/2012 Location lost : B016 (wrong information)	Display claim unsuccessful and required the claimant to provide correct data.		
3.	To update claim status of item using valid data	Status : Pending To Status : Saved	New status is successfully saved		
4.	To verify whether the lost item description match with the description from the claimant successfully	Description of claimant item: “Black prada wallet”	Successfully match the lost item description with claimant descriptions		
5.	Reject unverified claim	Item name : “White wallet”(do not match the lost item record)	System failed to find the matching item and disallow verification		

Admin Inventory & Storage Management Module

No	Objective	Test Data	Expected Result	Actual Result	Remarks/ Comments
1.	To organize the lost item in the storage using valid data.	Shelf’s category: Blue Phone Bin’s category: Red Clothes	Staff can find the organized lost item and give it to the rightful owner.		

		Room number: Room 01A			
2.	To organize the lost item in the storage using invalid data.	Shelf's category: Red Clothes (at the phone shelf) Bin's category: Blue Phone (in the clothes bin) Room number: Room 03B (in a wrong room number)	Staff will have a hard time to find the unorganized lost item to give to the rightful owner.		
3.	To update lost item's status using valid data	Lost item: Purple wallet Status: Claimed	Staff will know the lost items have been claimed.		
4.	To update lost item's status using invalid data	Lost item; Yellow jacket Status: Claimed (supposed to be unclaimed)	Staff will think the loss has been claimed without thinking and the item will probably be forgotten.		
5.	To look up the item with keywords or ID numbers using valid data.	Keyword: Book Or ID number: 782930183948	Staff will be able to look up what item the students are describing.		

Part 3.2 Software Configuration Management

Explore git